

Coding & Solution

Date	02 October 2026
Team ID	SWTID-2026-5625
Project Name	FitBuddy
Maximum Marks	

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/sharmiladharanidharan72007-create/FitBuddy-AI-Fitness
Programming Language(s)	Python, HTML5, CSS3
Framework(s) Used	FastAPI, Uvicorn, Jinja2 templates, SQLAlchemy (SQLite), Google Gemini AI (2.5 Pro / 2.5 Flash)
Key Features Implemented	• User profile input: name, user ID, age, weight, fitness goal, workout intensity and optional health information • Personalized 7-day workout plan (warm-up, cool-down, sets, reps, duration, rest) using Gemini AI • Personalized 7-day food plan (breakfast, lunch, evening snack, dinner) • Health-aware planning that adjusts exercises, intensity and rest • Feedback-based plan updates: Gemini AI modifies the existing plan • FastAPI backend with input validation and routing; SQLite storage of users, plans and feedback • Simple, user-friendly web interface (HTML, CSS, Jinja2); application prepared for local deployment
Pending / Incomplete Features	No major pending features. Cloud deployment is not yet done (runs locally).
Setup / Run Instructions	1. Clone: <code>git clone https://github.com/sharmiladharanidharan72007-create/FitBuddy-AI-Fitness.git</code> 2. Open the folder: <code>cd FitBuddy-AI-Fitness</code> 3. Create and activate a virtual environment: <code>python -m venv venv</code>, then <code>venv\Scripts\activate</code> (Windows) 4. Install packages: <code>pip install -r requirements.txt</code> 5. Create a <code>.env</code> file with <code>GOOGLE_API_KEY=YOUR_GEMINI_API_KEY</code> 6. Run: <code>uvicorn app.main:app --reload</code> 7. Open <code>http://127.0.0.1:8000</code> (API docs at <code>/docs</code>)

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

TEAM MEMBERS AND ROLES

1. Sharmila D – Team Lead

- Managed the overall project development.
- Handled project pre-requisites and topic links.
- Worked on the core functionalities of FitBuddy.
- Coordinated the team members and project activities.
- Prepared the application for local deployment.

2. Bhavana A – Backend Developer

- Worked on the project workflow.
- Developed and handled the FastAPI backend.
- Worked on routing and user-input handling.
- Tested the application during local deployment.

3. Rajashree K – AI/GenAI Developer

- Worked on Gemini AI integration.
- Researched the suitable Gemini model for the project.
- Worked on the main application logic.
- Contributed to the conclusion and AI-related functionality.

4. Siva Sundari V – Frontend/UI Developer

- Worked on the application architecture.
- Designed the user interface.
- Developed the frontend components.
- Focused on creating a simple and user-friendly interface.

5. Praveena K – Development and Template Developer

- Worked on the development environment setup.
- Developed dynamic templates using FastAPI and Jinja2.
- Worked on connecting the frontend templates with the backend.
- Supported application development and testing.